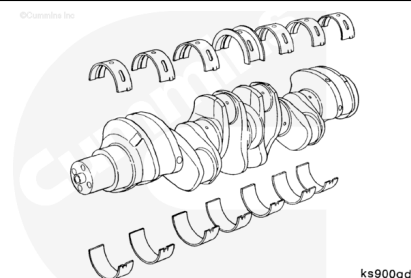


## Trainings ▾

### General Information

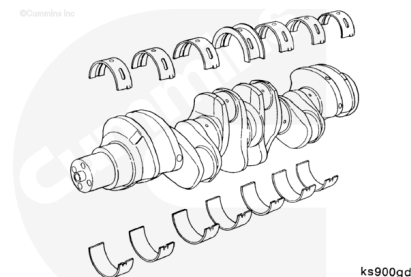
The C Series engine has seven main bearings. The lower bearing shells are all the same. The upper bearings do **not** have holes. All of the upper bearing shells are the same except for the Number 4 journal. The Number 4 journal uses a flanged upper bearing shell. The flanges on the bearing control the end thrust of the crankshaft.



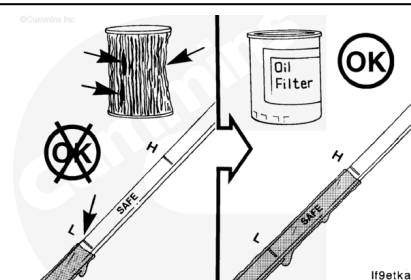
**Note : Oversize service main bearings are available for reground crankshafts.**

#### Main Bearing Sizes

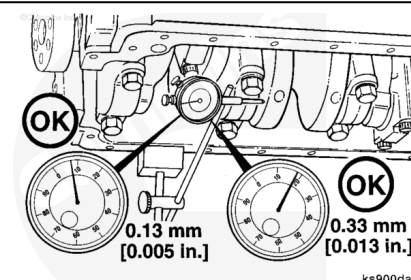
|          |                      |
|----------|----------------------|
| Standard |                      |
| Oversize | - 0.25 mm (0.009 in) |
|          | - 0.50 mm (0.019 in) |
|          | - 0.75 mm (0.029 in) |
|          | - 1.00 mm (0.039 in) |



Improper maintenance of the lubricating system is the primary cause of reduced main bearing life.



Increased end loading of the engine and its driven units at the front or rear of the engine can cause damage to the thrust bearings.



## Preparatory Steps

### **⚠ WARNING ⚠**

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

### **⚠ WARNING ⚠**

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

Drain the lubricating oil. Refer to Procedure 007-037 (</qs3/pubsys2/xml/en/procedures/100/100-007-037.html>).

Remove the lubricating oil pan and gasket. Refer to Procedure 007-025 (</qs3/pubsys2/xml/en/procedures/100/100-007-025.html>).

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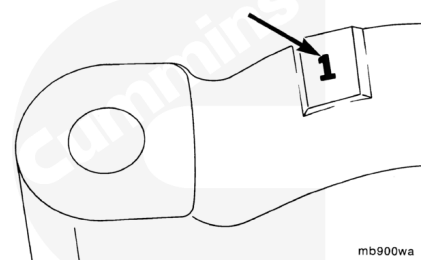


ck800wa

## Remove

**Note :** Before removing the main bearing caps, make sure that the caps are clearly marked for their location on the lubricating oil cooler side of the main bearing cap.

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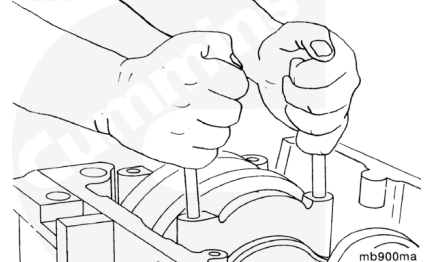


mb900wa

**Note :** Do not pry on the main caps to free them from the cylinder block.

Use two of the main bearing cap bolts to “wiggle” the main cap loose. Try **not** to damage the bolt threads.

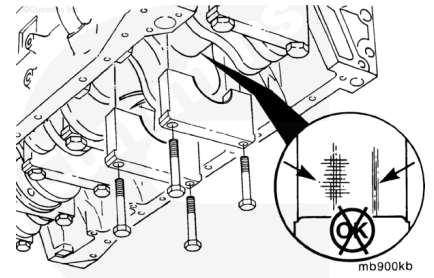
©Cummins Inc.



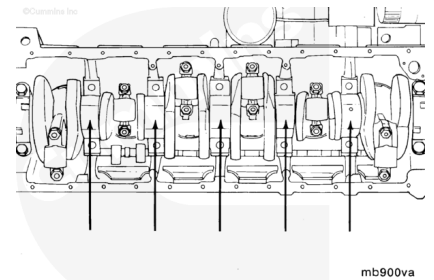
mb900ma

Remove the Number 2 and Number 3 main bearing caps; check the crankshaft journals for signs of overheating, deep scratches, or other damage. If there is no damage, there is no need to remove the other main bearing caps at this time.

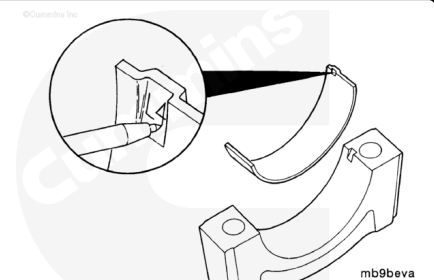
**Note :** The Number 4 main bearing cap has a thrust bearing.



If the main bearings are to be removed, remove all bearing caps except for cylinders Number 1 and 7.



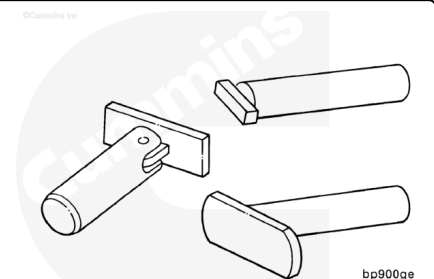
Mark the main bearing shell with the letter L (lower) and the journal number from which it was removed.



### ⚠ CAUTION ⚠

**Do not use a capscrew or any hard metal object; they can damage the crankshaft and cause serious engine damage.**

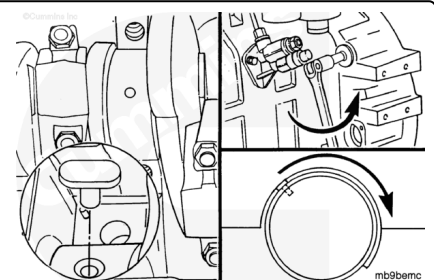
To remove the upper main bearing shell, install the main bearing rollout tool, Part Number 3823818, in the oil hole of the main bearing journal.



Using the engine barring tool, Part Number 3377371, rotate the crankshaft so the main bearing rollout tool, Part Number 3823818, pushes against the end of the bearing opposite the tang.

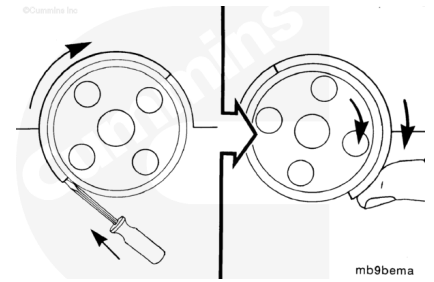
Continue rotating the engine to remove the bearing.

Follow this procedure to remove the other bearings.



### ⚠ CAUTION ⚠

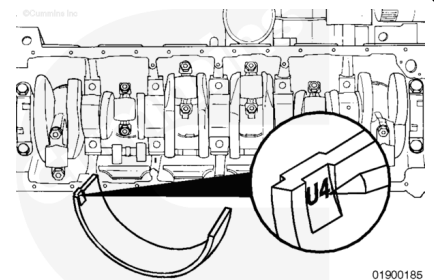
Use care so the screwdriver does not damage the crankshaft or cylinder block.



**Note :** The Number 1 main does **not** have a hole in the journal, so the bearing rollout tool can not be used to replace the bearing.

Use a flat-blade screwdriver. **Gently** bump the end of the bearing to loosen it from the block. Then, use finger-pressure against the bearing shell, and rotate the crankshaft to roll the bearing out.

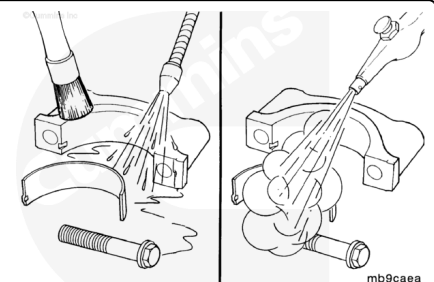
Mark the main bearing shell with the letter U (upper) and the journal number from which it was removed.



## Clean and Inspect for Reuse

### ⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



### ⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

### ⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

### ⚠ CAUTION ⚠

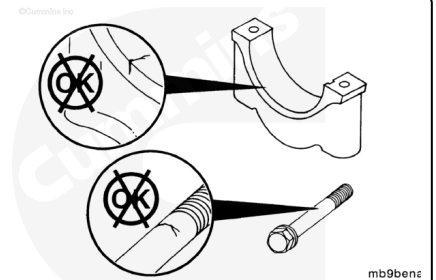
**Do not use a scraper or wire brush to clean the main bearing.  
Bearing damage can occur.**

Use solvent and a soft bristle brush to clean the cap, capscrews, and bearings.

Dry the main bearing caps, capscrews, and main bearings with compressed air.

Inspect the main cap and capscrews for damage.

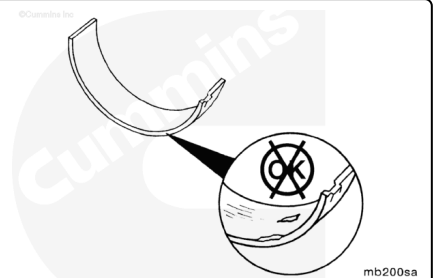
Inspect the bearing shells and the thrust bearings for nicks, scratches, or damage.



Inspect the main bearings for nicks, scratches, or damage.

Replace any bearing with the following damage:

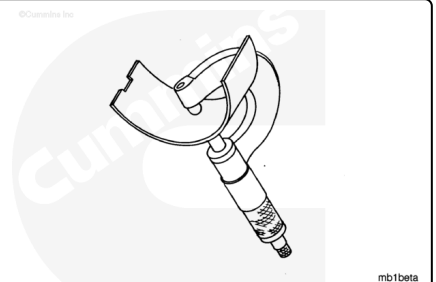
- Pitting
- Flaking
- Corrosion
- Lock Tang Damage
- Scratches (deep enough to be felt with a fingernail).



**Note :** For more detailed information on bearing damage, refer to Analysis and Prevention of Bearing Failures, Bulletin Number 3810387, or Parts Reuse Guidelines, Bulletin Number 3810303.

Use an outside diameter ball-tipped micrometer to measure the main bearing shell thickness.

| mm    |     | in     |
|-------|-----|--------|
| 3.446 | MIN | 0.1357 |
| 3.454 | MAX | 0.1360 |

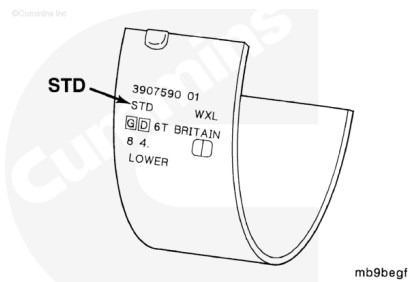


Discard any main bearing shell if it is below the minimum specification.

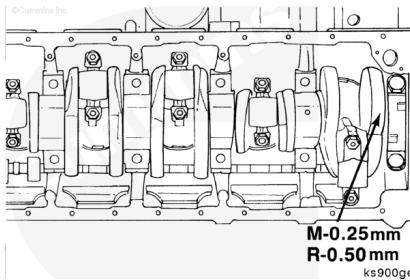
**Note :** The upper bearings contain two oil holes. The lower bearings do not.

The main bearings are marked on the back to indicate either standard (STD) or oversize (OS).

Determine the size of the bearing removed, and obtain the same size for installation.



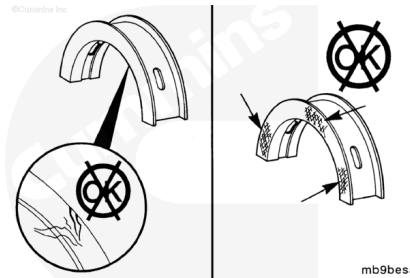
**Note :** Main bearing shells are available for crankshafts that are 0.50 mm [0.020 in] undersize. Crankshafts that are machined undersize in the connecting rod and main bearing journals are marked on the front counterweight. If the crankshaft is marked, check the bearing shell part number to make sure the correct bearing size is used.



### Thrust Bearing

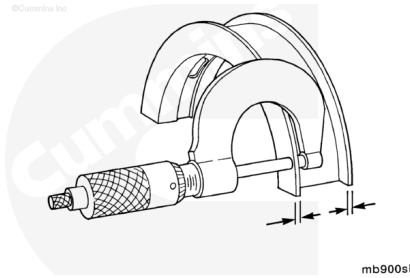
Inspect the thrust bearings for wear, cracks, and damage on the front and rear face.

Inspect the crankshaft journal contact area.



Measure the thrust bearing flange thickness. Measure both the front and rear flange of the thrust bearing.

| mm   |     | in    |
|------|-----|-------|
| 3.52 | MIN | 0.139 |
| 3.57 | MAX | 0.141 |

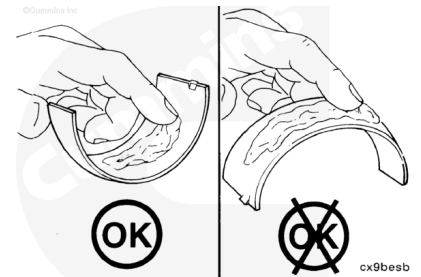


**Note :** The thrust bearings are available in 0.25-mm [0.010-in] and 0.50-mm [0.020-in] oversize thicknesses.

## Install

**⚠ CAUTION ⚠**

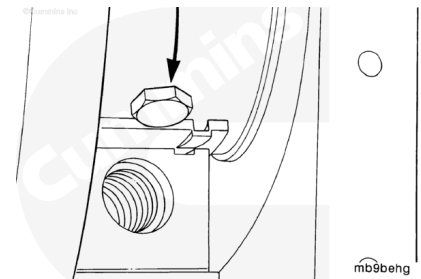
The cylinder block saddle and the main bearing cap mating surfaces must be clean and dry when the bearing shells are installed. Used bearings must be installed in their original location.



**Note :** Do not lubricate the back of the bearing shell.

Use clean engine oil to lubricate the upper main bearing shell to the crankshaft journal mating surface.

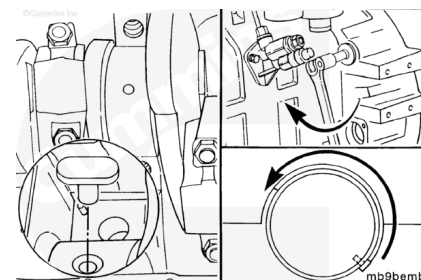
Position the new upper main bearing on the crankshaft and install as far as possible by hand.



**⚠ CAUTION ⚠**

**Make sure the pin does not slide under the bearing and damage the crankshaft.**

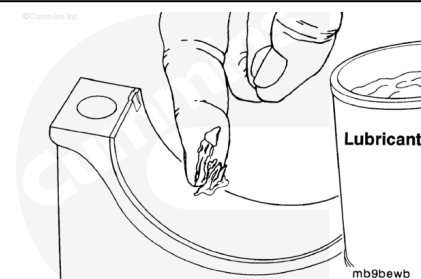
Use the main bearing rollout tool, Part Number 3823818, and the engine barring tool, Part Number 3377371, to finish pushing the bearing into position. Make sure the upper main bearing is aligned with the block and that the bearing tang aligns with the notch in the cylinder block.



Install the lower main bearings into the bearing caps.

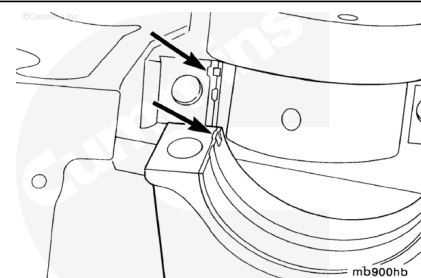
**Note :** Make sure the tang of the bearing shell is in the slot of the cap.

Use clean engine oil to lubricate the bearing shells.

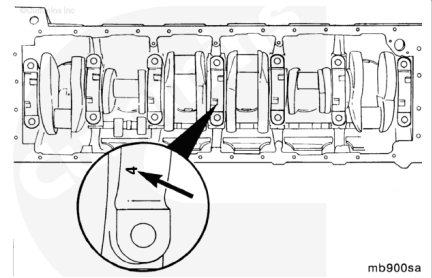


**Note :** The main bearing caps are numbered for location. Number 1 starts with the front of the block, and the numbers face the camshaft side of the engine.

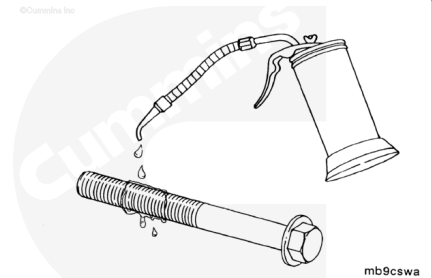
When correctly installed, the tang (slots) will both be on the same side of the cylinder block.



**Note :** The thrust bearing journal does not have locking tangs. Care must be taken to make sure that the number stamped on the main bearing cap is positioned facing the camshaft side of the engine.



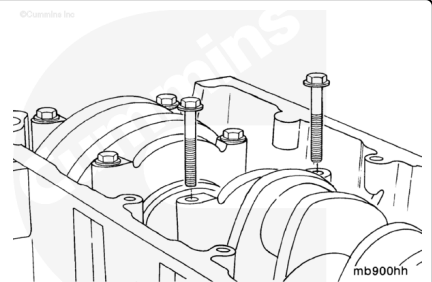
Use clean engine oil to lubricate the capscrew threads.  
Drain the excess oil from the capscrews before installing them.



Tap the main bearing caps gently into position.

**Note :** Make sure the bearing does not become dislodged from the cap.

When seated, the main bearing cap bolts can be threaded by hand.

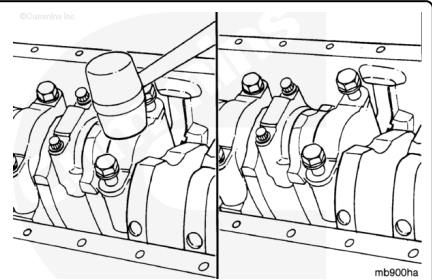


Install a main bearing cap after each upper bearing is installed to keep the bearing in place while the other uppers are installed.

**Note :** Do not tighten the main bearing cap capscrews to the final torque value at this time.

Tighten the capscrews to an initial torque.

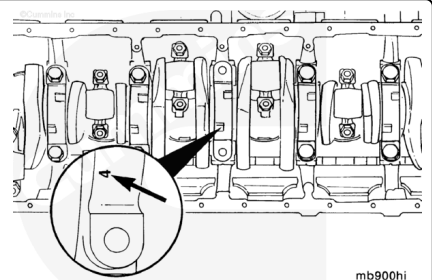
**Torque Value:** 50 N·m [ 37 ft·lb ]



**Note :** The thrust journal does not have tangs or slots.

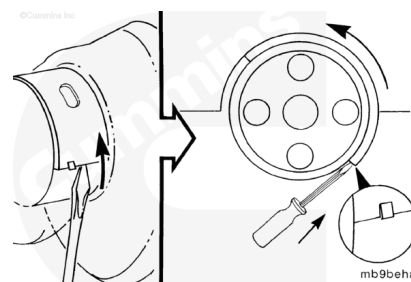
**Note :** Make sure the number stamped on the main bearing cap is positioned facing the camshaft side of the cylinder block.

Push the crankshaft toward the center of its end thrust position to allow clearance for the thrust bearing.



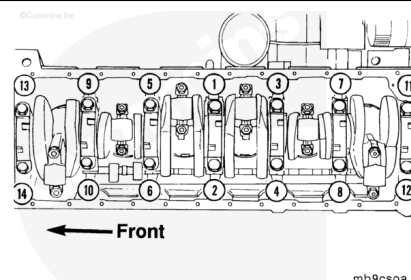


Since the main bearing rollout tool, Part Number 3323818, **cannot** be used for the Number 1 main bearing, use a blunt object or screwdriver to push the bearing into position as the crankshaft is rotated.



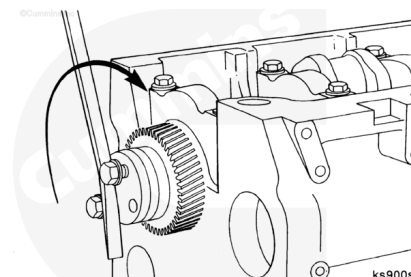
Tighten the capscrews evenly following the sequence shown.

|                      |        |                   |
|----------------------|--------|-------------------|
| <b>Torque Value:</b> | Step 1 | 50 n.m [37 ft-lb] |
|                      | Step 2 | 95 n.m [70 ft-lb] |
|                      | Step 3 | 60° turn.         |



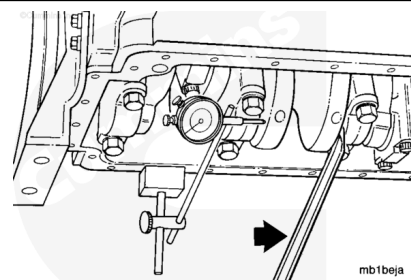
**Note :** The crankshaft must rotate freely after the main bearings have been tightened.

Check the main bearing installation and the size of the main bearings if the crankshaft does **not** rotate freely.

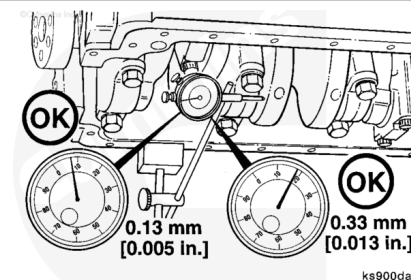


Measure the end clearance of the crankshaft as follows:

- Install a dial indicator onto the oil pan flange.
- Put the tip of the gauge against the crankshaft counterweight.
- Push the crankshaft toward the rear of the cylinder block.



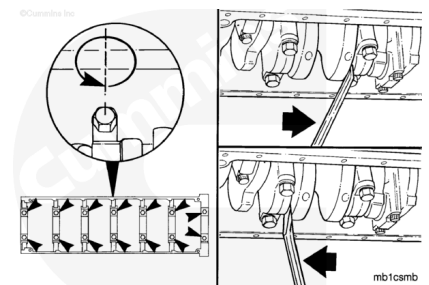
- Set the dial indicator at zero.
- Push the crankshaft toward the front of the cylinder block.



If the end clearance is less than 0.127 mm [0.005 in], do the following:

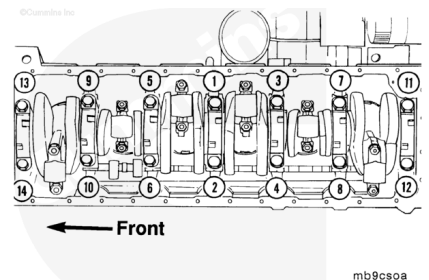
- Loosen the main bearing capscrews one turn.

- Push the crankshaft toward the front and then toward the rear of the cylinder block.



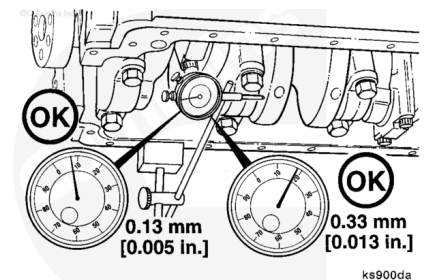
Tighten the main bearing capscrews in the sequence shown.

|                      |        |                   |
|----------------------|--------|-------------------|
| <b>Torque Value:</b> | Step 1 | 50 n.m [37 ft-lb] |
|                      | Step 2 | 95 n.m [70 ft-lb] |
|                      | Step 3 | 60° turn.         |



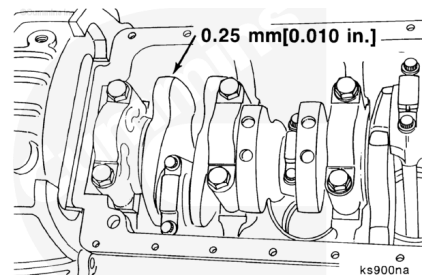
Measure the crankshaft end clearance.

| mm    |     | in    |
|-------|-----|-------|
| 0.127 | MIN | 0.005 |
| 0.330 | MAX | 0.013 |

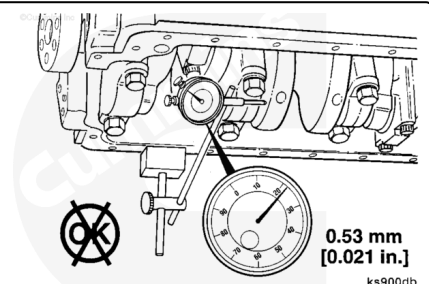


**Note :** Crankshafts that have been reground on the thrust bearing surfaces are marked for oversize thrust bearings on the rear crankshaft counterweight. If the crankshaft counterweight is marked, check the thrust bearing part number to make sure the correct bearing size is used.

Example: 0.010 = 0.25 mm [0.010 in]



If the crankshaft end play is more than 0.330 mm [0.013 in] new, or 0.533 mm [0.021 in] used, the crankshaft **must** be removed from the engine and replaced.



## Finishing Steps

Install the oil suction tube. Refer to Procedure 007-035 (</qs3/pubsys2/xml/en/procedures/100/100-007-035.html>).

Install the oil pan. Refer to Procedure 007-025 (</qs3/pubsys2/xml/en/procedures/100/100-007-025.html>).

Fill the engine with lubricating oil. Refer to Procedure 007-037 (</qs3/pubsys2/xml/en/procedures/100/100-007-037.html>).

**Note :** Use only Cummins™ approved lubricating oil for natural gas engines to prevent premature engine damage.

Operate the engine until the coolant temperature reaches 82°C [180°F], and check for lubricating oil leaks.

